CHAPTER

9

Missing Persons and Disaster Victim Identification Efforts

Kinship analysis and parentage testing typically involve one sample being compared to another sample or to a few samples with a specific relationship question in mind. This chapter will review efforts with missing persons investigations and disaster victim identification situations where a set of remains is being compared to many different samples, which introduces another level of complexity beyond the challenges of kinship analysis.

MISSING PERSONS INVESTIGATIONS

An estimated 40,000 unidentified human remains have been recovered and are currently located in medical examiner and coroners' offices around the United States (Ritter 2007). Every year in the United States, tens of thousands of people are reported "missing," often under suspious circumstances. While some of these missing persons are later located alive through law enforcement efforts, many become unidentified human remains that resulted from criminal activity, such as rape and murder. Knowledge of who the victim is can help solve these crimes and bring closure to families of the missing.

There are three categories of samples associated with missing persons cases: direct reference samples, family reference samples, and unidentified human remains (UHR) samples. The UHR samples are generally skeletal remains (bones), teeth, or tissue. Much of the data from missing person investigations is in the form of mitochondrial DNA sequences since this information can be successfully recovered from highly degraded samples. Mitochondrial DNA also enables access to a larger number of reference samples from maternal relatives of a victim (see Chapter 14).

Some possible direct reference samples include medical samples from the missing individual, such as a newborn screening bloodspot or a biopsy sample. Personal effects, such as a toothbrush or hairbrush, may also provide direct reference samples. Family reference samples can be buccal swabs from close biological relatives, such as parents, children, or

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siblings of the missing individual. More distant relatives, such as maternal aunts, maternal or paternal uncles, or maternal or paternal cousins, can also be useful if mitochondrial or Y-chromosome DNA testing are performed. The combined evaluation of samples from more than one close relative can help provide greater confidence in such kinship analyses.

DNA databases can play an important role in helping to identify missing individuals over time. When a family member goes missing, DNA samples can be obtained from direct reference samples or biological relatives. DNA profiles from these samples would then be uploaded to the database and searched against DNA profiles from unidentified human remains in an effort to make an association to a missing individual. Many states within the United States and nations around the world are beginning to establish missing persons databases to enable matching of recovered remains to their family members. The National DNA Index System (NDIS) discussed in Chapter 8 also contains indices to help with missing persons investigations. Figure 9.1 illustrates these indices and purposes of searches between them.

In an effort to help make connections between family members and their missing relatives, in 2007 the U.S. government established a National Missing and Unidentified Persons System (NamUs) website at http://www.namus.gov/. NamUs is composed of two sets of records—a missing persons database (http://www.findthemissing.org) and an unidentified persons database (https://identifyus.org/)—that are cross-matched against one another. The missing persons database contains information that can be entered by the general public regarding an individual who has gone missing. Only medical examiners and coroners can enter information in the unidentified persons database, which describes bodies found but not yet identified. However, as noted on the NamUs website, anyone can search the unidentified persons database using characteristics such as sex, race, distinct body features, and even dental information. The NamUs website provides the capability to print a missing persons poster with the missing individual's name, his or her photo, and contact information for investigators or family members.

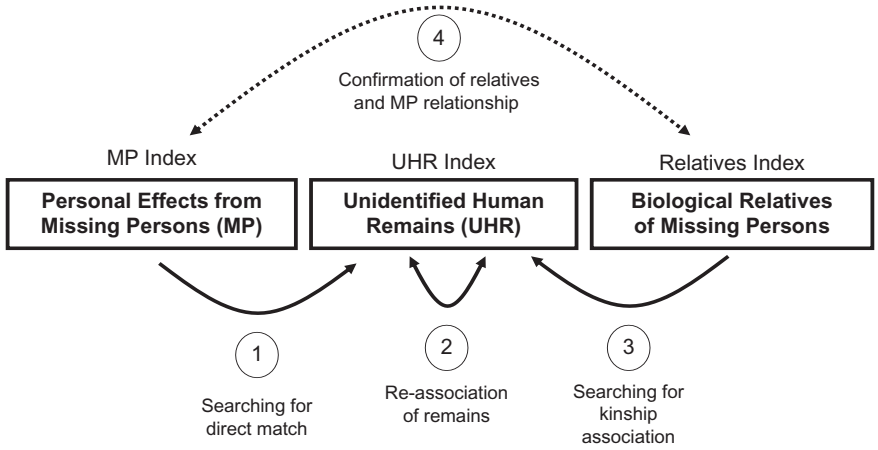


FIGURE 9.1 Searches conducted with missing persons DNA databases.

As of December 2010, NamUs listed 7538 unidentified human remains cases, of which 7214 (almost 96%) were still open. Of the 324 cases closed, 25 (or about 8%) had been aided by NamUs. At the same time, the missing persons database listed 5889 cases with 4912 (just over 83%) still open. Of the 977 cases closed, 44 (4.5%) had been aided by the NamUs website.

DISASTER VICTIM IDENTIFICATION (DVI)

Mass diasters, whether natural or man-made, can involve loss of life for many victims of the tragedy. Efforts to identify these victims are referred to as *disaster victim identification*, or DVI.

In the United States, DNA testing has now become routine and expected in disaster victim identification in the event of a plane crash, large fire, or terrorist attack. Military casualties are also identified through STR typing or mitochondrial DNA sequencing by the Armed Forces DNA Identification Laboratory (AFDIL). All airplane crashes within the United States are examined by the National Transportation Safety Board (http://www.ntsb.gov), which often contracts with AFDIL to identify the air crash victims through DNA testing as part of the investigation.

Often mass disasters leave human remains that are literally in pieces or burned beyond recognition. In some cases it is possible to visually identify a victim, but body parts can be separated from one another and the remains comingled making identification without DNA techniques virtually impossible. The use of fingerprints and dental records (odontology) still plays an important role in victim identification but these modalities obviously require a finger or an intact skull or jawbone along with previously archived fingerprint and dental records.

DNA testing has a major advantage in that it can be used to identify each and every portion of the remains recovered from the disaster site, provided (1) that there is sufficient intact DNA present to obtain a DNA type and (2) a reference sample is available for comparison purposes from a surviving family member or some verifiable personal item containing biological material. Personal items from the deceased including toothbrushes, combs, razors, or even dirty laundry can provide biological material to generate a reference DNA type for the victim. The direct comparison of DNA results from disaster victim remains to DNA recovered from personal items (Figure 9.2a) represents the easiest way to obtain a match—and hence an identification—provided it is possible to verify the source (e.g., the toothbrush was not used by some other household member). The use of DNA from biological relatives (Figure 9.2b) necessitates the added complexity of kinship analysis similar to that employed for paternity or reverse parentage testing.

DVI always involves comparison of post-mortem (PM) and ante-mortem (AM) data. PM data are generated from the recovered human remains, which may be highly fragmented depending on the type of disaster. AM data come from either direct reference samples (e.g., toothbrushes or razors known to belong to the victim) or kinship comparisons to biological relatives (e.g., parent, child, or sibling).

DVI is much more complicated than parentage testing because there are so many more comparisons that are being made depending on the number of victims involved. In addition, the quality of the recovered human remains may be compromised depending on the type of disaster and thus partial or mixed DNA profiles may result. DNA statistics for DVI work are usually best represented with likelihood ratios because this permits DNA results

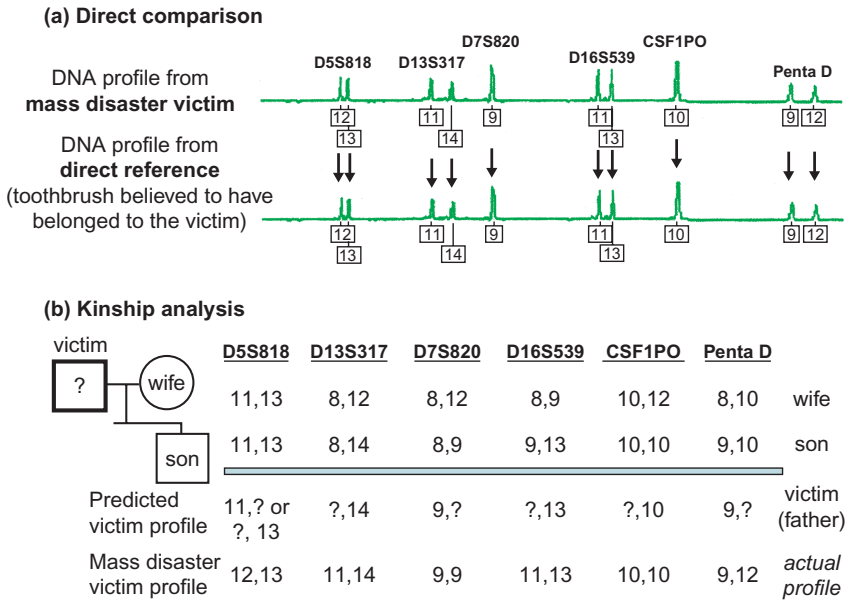


FIGURE 9.2 Example demonstrating the use of reference samples in mass disaster victim identification using DNA typing. (a) Direct comparison involves analysis of a direct reference sample from some kind of personal effect of the victim. (b) Kinship analysis utilizes close biological relatives, such as those illustrated in Figure 9.3, to reconstruct a victim's DNA profile.

to be combined between multiple genetic systems as well as other non-DNA evidence using Bayesian statistics.

DNA testing has been used to help identify victims of numerous airline crashes, the victims of terrorist attacks, recovered remains from mass graves, and in a more limited fashion in natural disasters like the Southeast Asia Tsunami in December 2004 and Hurricane Katrina that struck New Orleans in August 2005.

Several documents have been published with lessons learned from previous DVI work (AABB 2010, Budowle et al. 2005, Lessig et al. 2011, NIJ 2006). The International Society of Forensic Genetics (ISFG) published 12 recommendations for DVI work (Prinz et al. 2007) (D.N.A. Box 9.1).

ISSUES FACED DURING DISASTER VICTIM IDENTIFICATION EFFORTS

There are several important aspects of mass fatality incidents that will be discussed prior to moving into examples of victim identification efforts through DNA testing in recent mass disasters. The areas include collection of reference samples and federal assistance programs. The National Institute of Justice has published two documents to provide additional information on mass disaster investigations: *Mass Fatality Incidents: A Guide for Human Forensic*

D.N.A. BOX 9.1

ISFG RECOMMENDATIONS ON DISASTER VICTIM IDENTIFICATION

In the inaugural issue of the journal *Forensic Science International: Genetics*, the ISFG DNA Commission published 12 recommendations on disaster victim identification (DVI) using DNA (Prinz et al. 2007):

- **Recommendation #1:** Every forensic DNA laboratory should make an effort to contact the relevant authority dealing with emergency response and establish involvement in a possible mass fatality preparedness plan. Policy decisions about sample collection, scope and final goals of the effort will affect the victims' families and the work stream and should be decided as early as possible.
- **Recommendation #2:** The internal response plan needs to address throughput capacity, sample tracking, and must have names of supervisors responsible for different tasks that are updated as personnel changes.
- **Recommendation #3:** Several sample types (see Table 1 in Prinz et al. 2007) for DNA testing should be taken at the earliest possible stage of the investigation provided material traceability is guaranteed. Samples must be collected from each body or recognizable body part, even if identity is already established. Proper storage must be assured.
- **Recommendation #4:** Multiple direct references and samples from first-degree relatives should be collected for each missing person. Scientists with a background in genetics should be available for training or for consultations in the family liaison group.
- **Recommendation #5:** DVI DNA testing should only be performed by laboratories with demonstrated successful capabilities and continuous experience with these specified sample types.
- **Recommendation #6:** The set of loci to be analyzed has to be identified as soon as possible in concordance with the scientific community in the countries mostly involved. A minimum of 12 independent loci should be selected as standard set, but an even greater number of loci is preferred.
- **Recommendation #7:** All allele calls and all candidate matches have to be reviewed thoroughly. Composite DNA profiles can be generated if derived from the same specimen and consistent for overlapping loci. The duplication policy should consider the logistics and circumstances of the mass fatality incident.
- **Recommendation #8:** If the standard autosomal STR typing fails to give sufficient information, additional typing system such as mtDNA, Y-chromosomal STRs, or SNP markers may be used in selected cases.
- **Recommendation #9:** A centralized database is required for all data comparison. Electronic upload is recommended to avoid transcription errors.
- **Recommendation #10:** Especially if multiple family members are involved, DNA-based identification should whenever possible be anchored by anthropological and/or circumstantial data, a second identification modality, or multiple DNA references.

D.N.A. BOX 9.1 (*cont'd*)

- **Recommendation #11:** In DVI work, DNA statistics are best represented as likelihood ratios that permit DNA results to be combined among multiple genetic systems or with other non-DNA evidence. Likelihood ratio thresholds should be determined for when DNA data alone can suffice for an identification; this will be based on the size and circumstances (e.g. closed versus open) of the event. All evidence and/or circumstances should be checked in making an identification, even if DNA provides the primary or sole evidentiary factor.
- **Recommendation #12:** The preparedness plan of the laboratory needs to include policies for family notification, long-term sample disposition, and data archiving.

***Source*:**
Prinz, M., et al. (2007). DNA Commission of the International Society of Forensic Genetics (ISFG): Recommendations regarding the role of forensic genetics for disaster victim identification (DVI). Forensic Science International: Genetics, *1, 3–12.*

Identification (prepared by the National Center for Forensic Science, NIJ 2005) and *Lessons Learned from 9/11: DNA Identification in Mass Fatality Incidents* (prepared by the World Trade Center Kinship and Data Analysis Panel, NIJ 2006).

Collection of Reference Samples

In order to be able to identify victims of mass fatality incidents, reference samples are needed in order to sort out DNA profiles obtained from recovered remains. If possible, it is preferable to obtain personal effects that enable a direct match to a victim (Figure 9.2a). These personal effects may be in the form of a used razor, hairbrush, toothbrush, dirty laundry, or other items that were handled solely by the victim and from which usable biological material may be recovered to generate a DNA profile.

Living biological relatives can also provide needed and valuable reference samples. Immediate relatives including siblings, parents, and children are the most effective indirect reference samples (Figure 9.3). More extended family members can provide helpful samples though if mitochondrial DNA or Y-chromosome testing is performed.

Kinship samples can also help confirm the validity of personal effects received for a missing individual. Often a lengthy and complicated administrative review of DNA results and reference sample chain-of-custody is needed to verify both direct reference samples from personal effects and indirect references from kinship samples to enable confidence in reporting DNA identifications (see Hennessey 2002).

Federal Assistance in Disaster Situations

The Disaster Mortuary Operational Response Team (DMORT) is a federally funded group of professionals with experience in disaster victim identification that becomes activated

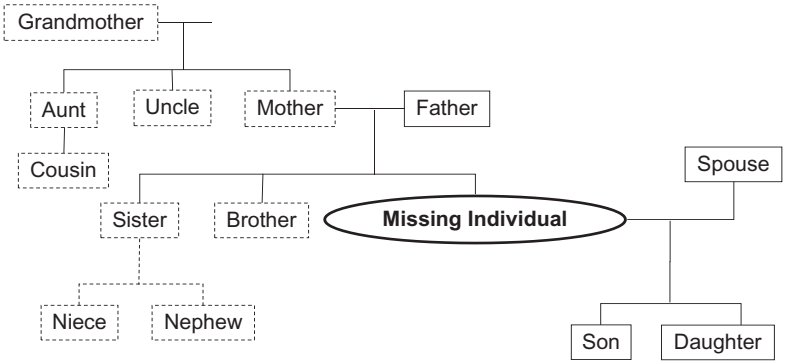


FIGURE 9.3 Direct biological relatives that can provide valuable reference samples for aiding identification of the missing individual. Ideally samples are available from multiple relatives to help establish robust kinship. A sample from a spouse is only valuable in connection with a child's sample in order to determine the expected alleles coming from the missing individual. Of course, more extended family members, with direct maternal or paternal linkage, can provide helpful samples when mitochondrial DNA or Y-chromosome testing is performed. Samples from extended family members that would be valuable for the maternally transmitted mitochondrial DNA are indicated in dashed boxes.

in response to a major disaster in the United States (see http://www.dmort.org). DMORT is part of the National Disaster Medical System, U.S. Department of Health and Human Services, and can be activated by one of four methods: (1) a request for assistance from a local official through the Federal Emergency Management Agency (FEMA) when a federal disaster has been declared, (2) a request for assistance from the National Transportation Safety Board when a passenger aircraft accident occurs, (3) under the U.S. Public Health Act to support a state or locality that cannot provide the necessary response to a disaster, and (4) when requested by a federal agency, such as the FBI, to provide disaster victim identification.

Usually within 24 to 48 hours after the disaster, the DMORT team is set up to provide professional personnel and technical support and assistance to the local medical examiner or coroner in forensic services and victim identification. The team is composed of forensic pathologists, forensic anthropologists, forensic dentists, medical investigators, funeral directors, and other technical support staff.

Dental records and X-rays along with fingerprints are normally the primary methods used in victim identification. DNA will be used as a last resort and only after all conventional means of identification are exhausted. DMORT does not perform DNA testing but rather gathers samples for future DNA testing, often by a laboratory such as AFDIL.

FEMA provides funding for the DNA identification effort along with other aspects of a disaster investigation after an official declaration of a major disaster or state of emergency has been declared by the President of the United States following a request for such a declaration from the governor of the affected state. This "state of emergency" declaration then activates a number of federal programs to assist in the response and recovery effort (see http://www.fema.gov).

Why Identify Remains of Victims?

Perhaps some wonder why the limited and precious resources of law enforcement and forensic laboratories are devoted to the challenging effort of identifying the remains of disaster victims. Although a variety of reasons can be cited, there are two primary reasons that impact living family members. First, in many jurisdictions, some form of identification is required before a death certificate can be issued that enables remaining family members to collect on life insurance policies. Second, the living family members may want the remains of their loved one returned to bring closure to the tragedy and to provide a proper burial and memorial service for their relative. As noted by Ballantyne (1997), there seems to be an overwhelming desire by many relatives to retrieve even the most miniscule tissue sample of a loved one.

Challenges with DNA Identification of Victims

Occasionally there are no known or living biological relatives or all immediate members of a family are among the victims making it difficult to associate the remains or to connect them to a valid reference sample. An additional challenge can come with simply locating surviving family members or communicating with them in the case of an international disaster.

Unfortunately, there can be challenges in dealing with family members of disaster victims such as family disputes (e.g., feuding family members fighting over who is entitled to the recovered remains) or discovery of illegitimate relationships when biological nonpaternity is demonstrated for someone who previously thought that they were the father or the child of a victim. Care must be taken by the laboratory director or other laboratory personnel who may interact directly with the families of victims to be sensitive to their grieving processes.

Collection of biological material from a disaster site is sometimes anything but trivial. For example, if a plane goes down at sea (e.g., Swissair Flight 111, see below), especially in deep water, then recovery of the remains can be quite challenging. By the very nature of the disaster, there is typically damage done to the biological samples and hence the DNA molecules contained therein. Extreme environmental conditions both during and after the disaster impact the quality of the recovered remains so that the DNA may be degraded, which can complicate DNA analysis and interpretation by the laboratory. As described in Chapter 10, degraded DNA gives rise to partial profiles, which then lower the statistics of a match because not all loci tested can be reported.

Trauma to Laboratory Personnel

Being exposed to large numbers of badly damaged human remains can have a psychological impact on laboratory personnel. In addition, loved ones may be among those who died making it difficult to cope with the rigors of careful laboratory work. Strain can be placed on the laboratory by political officials and the news media to produce results rapidly and to give regular updates of progress.

EARLY EXAMPLES OF DVI USING DNA

In the following pages, several mass disaster situations are highlighted where STR typing proved to be a valuable means for identifying human remains from burn victims, airplane crashes, terrorist acts, and mass grave sites. In these cases, STR typing was successfully performed in spite of heavy damage inflicted by a high-temperature fire (Branch Davidian compound in Waco, Texas, April 1993) or severe water damage (airplane crash of Swissair Flight 111 near Nova Scotia, September 1998). Identification of the victims of mass disasters, such as the terrorist attacks of September 11, 2001 on the World Trade Center Twin Towers, can require application of innovative biology, technology, and genetics.

Waco Branch Davidian Fire

On April 19, 1993, the Mount Carmel Branch Davidian compound in Waco, Texas burned during a raid by FBI agents. Over 80 individuals died and their remains were severely damaged following a high-temperature fire. While approximately half of these individuals could be identified by dental or fingerprint comparison and anthropological and pathological findings, the rest had to be identified based on information that could only be provided by DNA analysis. This was the first mass disaster investigation where DNA analysis with STR markers was used.

The Armed Forces DNA Identification Laboratory (AFDIL) and the Forensic Science Service (FSS) in England analyzed these samples by examining a variety of DNA markers including HLA DQ alpha, AmpliType PM, D1S80, amelogenin sex-typing, mitochondrial DNA sequencing, and four STR loci (TH01, F13A1, FES/FPS, VWA). Without the use of polymerase chain reaction (PCR)-based DNA typing procedures, specifically STR markers, approximately half of the individuals who perished in the Mount Carmel Compound of Branch Davidians would not have been identified.

AFDIL received 242 samples from the Mount Carmel Branch Davidian compound representing 82 sets of human remains (DiZinno et al. 1994). Blood-stain cards from living relatives provided reference samples for the unknowns. When usable tissue from the badly burned bodies was not available, portions of rib bones were removed and the DNA was extracted.

Body identifications were made by matching observed sample genotypes with predicted possible genotypes obtained from using results of relatives' reference blood samples and information gathered from family trees. This approach is basically a reverse parentage analysis where the parent genotypes are used to predict the child's genotype. A total of 26 positive identifications were made using the family tree matching approach (Clayton et al. 1995). A shortage of relatives prevented the identification of the other bodies. These results highlight the need for reference samples in order to take full advantage of DNA testing in mass disaster situations (Ballantyne 1997).

Spitsbergen and TWA Flight 800 Airplane Crashes

Two airplane crashes in 1996 helped demonstrate the value of DNA analysis in victim identification. On August 26, 1996, an airplane carrying 64 Russian and 77 Ukrainian

individuals crashed into a mountain near Spitsbergen killing all 128 passengers and 13 crewmembers. The establishment of victim identity took only 20 days due to rapid analysis of samples in the University of Oslo (Norway) Institute of Forensic Medicine.

DNA profiles comprised of three STRs and five minisatellite loci enabled associating 257 body parts to 141 different individuals. Of these assortments, 139 were identified based on available reference samples submitted by 154 relatives (Olaisen et al. 1997). In this particular case, the rapid recovery of samples from a frigid environment (≈0°C) led to 100% success in producing DNA profiles from the remains. The cost of the DNA typing portion of the investigation was approximately 3% to 5% of the total cost of the entire operation.

Shortly after take-off on July 17, 1996, TWA Flight 800, a Boeing 747 en route to Paris, blew up in the sky above Long Island, New York, killing all 230 passengers and crew. Their fragmented remains were also identified with STR typing results (Ballantyne 1997). These early successes with DNA testing in aiding airline disaster victim identification paved the way for more recent uses.

Swissair Flight 111

On the evening of September 2, 1998, while en route to Geneva, Switzerland, from New York City, Swissair Flight 111 crashed into the Atlantic Ocean not far from Halifax, Nova Scotia. All 229 people on board (214 passengers and 15 crewmembers) were killed. The plane went down about 10 kilometers from land, requiring the wreckage to be raised from a depth of more than 60 meters (≈180 feet) of water.

Over the next few weeks, a large task force of investigators collected human remains from the crash scene. In addition to the proof-of-death and familial closure reasons discussed above, these remains were carefully collected and subsequently identified as an essential part of the crash investigation. Without a mechanical reason for the plane falling from the sky, criminal activity was a possibility. The plane's manifest listed 229 people on board. But were they all who they claimed to be? Any discrepancy from that number could be a sign of terrorist activity. A missing individual or an extra passenger who could not be accounted for might have been a terrorist with a bomb.

An important reason for identifying the victims of any mass disaster is to bring closure to the living relatives. If the remains of their loved ones can be identified, then something can be given back to the living relatives for burial and memorial purposes. However, one of the challenges for identifying the remains of airline crashes is that entire families often travel together. In this case, closely related individuals have to be distinguished from each other and sometimes without the benefit of a living relative to act as a reference sample.

A number of methods were used to identify the victims, including fingerprints, dental records, X-ray evaluation, and DNA testing. Only one body was intact enough for visual identification. A total of 147 victims could be identified by means other than DNA. For example, 1020 fingers were recovered from the crash site. However, these fingers allowed only 43 victims to be identified based on their fingerprints because only a small percentage of the victims had fingerprint records that could be located. Police visited the homes of the victims and tried to recover latent fingerprints from objects that they may have handled. These efforts led to the recovery of over 200 latent prints that were used to identify 33 of 43 victims mentioned above. An effective method of initial identification involved dental records, which were used to

positively identify 102 of the victims and enable a certificate of death to be issued. Dental comparisons provided the fastest identification when reference samples were available.

Concurrent to other efforts to identify the crash victims, DNA testing was performed by the Royal Canadian Mounted Police (RCMP). DNA analysis was performed by four RCMP laboratories from across Canada and the Ontario Provincial Forensic Laboratory, each contributing a vital and specific subset of data. The DNA identification process was coordinated by the DNA Methods and Database section in Ottawa. This team of more than 50 DNA scientists consisted of members of the Biology sections of the RCMP Forensic Laboratories located in Halifax, Regina, Vancouver, and Ottawa and the Centre for Forensic Sciences in Toronto. DNA typing with the STR markers described in this book was used to help identify all 229 people on board Swissair Flight 111 (Leclair et al. 2004). In every case where other forms of identification were performed, DNA analysis helped confirm and support those results.

Two separate identification issues were addressed by DNA testing. First, recovered human remains showing identical STR genotypes were associated. Second, each passenger was identified through comparisons of human remains with the reference samples isolated from

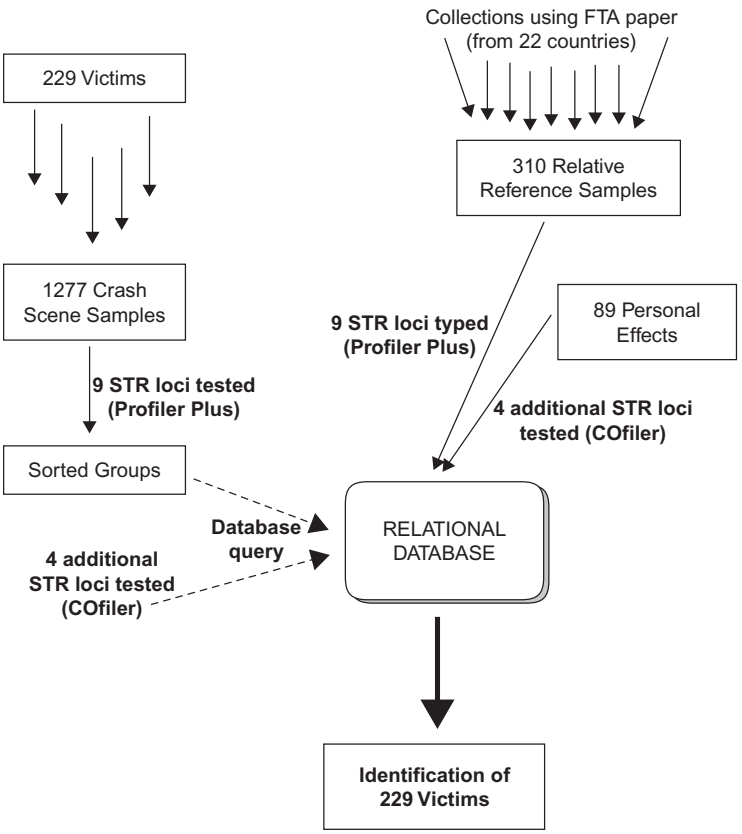


FIGURE 9.4 Strategy for disaster victim identification in the Swissair Flight 111 crash scene investigation. Most samples were tested with 9 STRs and amelogenin using the AmpFlSTR Profiler Plus kit and then 4 additional STRs were added with the COfiler kit as needed to obtain a higher power of discrimination between closely related individuals. The relational database compared genotypes of 1277 crash scene samples to genotypes of all 229 victims and family relatives, a total of 71,490 genotype comparisons (Leclair et al. 2004).

personal effects of the victims or reference blood samples submitted by relatives of the victims. In many cases, DNA analysis was the only means by which the samples could be positively identified.

Over 2400 human remains were recovered from the crash site of which 1277 were analyzed by DNA testing (Figure 9.4). These samples were analyzed along with 310 reference samples from relatives that were submitted on FTA paper blood cards (see Chapter 1) to be genotyped and used in a relational database. In addition, 89 personal effects, such as toothbrushes and hair from combs, were taken from the homes of 47 victims because no relatives were available to serve as a reference. One of the challenges of collecting the reference samples was the fact that the living relatives were from 21 different countries. The FTA kit enabled rapid blood collection and room temperature delivery of the reference samples and aided the successful completion of the investigation.

The AmpFlSTR Profiler Plus STR markers D3S1358, VWA, FGA, D8S1179, D21S11, D18S51, D5S818, D13S317, and D7S820 were the primary means of identifying the remains although additional STR loci TH01, TPOX, CSF1PO, and D16S539 from the AmpFlSTR COfiler kit were used to gain a higher power of discrimination (Frégeau et al. 2000). COfiler amplifications were performed on 118 crash scene remains and 129 "known" reference comparison samples. The crash scene samples were analyzed in approximately one week using either the nine STRs from Profiler Plus or all 13 STRs from Profiler Plus and COfiler. The challenge of making appropriate associations between the samples took a little longer. Genotypes of each of the 229 victims were compared to genotypes of all other victims and family relatives, which represented 71,490 genotype comparisons (Leclair et al. 2000, Leclair et al. 2004). The comparison of reference sample genotypes to profiles from the recovered remains involved over 180,000 comparisons because more known sample genotypes existed for cross-comparison purposes.

Traditional parentage trios with both living parents were encountered for only 25% of the 229 victims. Even more challenging was the fact that 43 families of 2 to 5 individuals were present among the victims. A pair of identical twins was also present on the plane and could not be individually identified with DNA testing. Nevertheless, the DNA testing led to confident kinship analysis in the case of 218 victims for whom reference samples from close relatives or personal effects were submitted (Leclair et al. 2000, Leclair et al. 2004).

The efforts of the RCMP demonstrated a successful model for how a mass disaster investigation should be conducted. Tremendous cooperation is required from forensic laboratories, law enforcement personnel, and family members of victims, often from a number of countries, in order to successfully identify the victims of mass fatality incidents.

The need for readily available reference samples was also highlighted by this investigation. In fact, a formal recommendation was made from the RCMP to the Canadian Transport Safety Board for all airline personnel and any private citizens who are frequent fliers to have fingerprints and DNA samples made available for identification purposes if ever the need arises. These records cannot be stored by the police but rather could be maintained by the airline or stored in an individual's safety deposit box.

Of the 229 victims of Swissair Flight 111, all 229 were positively identified, an astounding success compared to previously used conventional identification techniques and one that would not have been possible without the power of DNA typing with STR markers.

DNA IDENTIFICATION WORK WITH SEPTEMBER 11, 2001 VICTIMS

The terrorist attacks against the United States on September 11, 2001 left over 3000 victims in three different locations: the Pentagon in Washington, DC, a field near Shanksville (Somerset County), Pennsylvania, and the Twin Towers of the World Trade Center in New York City. Several teams of forensic scientists were involved in the DNA analysis of these mass disasters.

The Pentagon Site

Hijacked by five terrorists on the morning of September 11, 2001, American Airlines Flight 77 crashed into the Pentagon shortly after 9:40 a.m. killing all 59 passengers and crew on board, the 5 terrorists, and 124 Pentagon occupants. The remains, ranging in size from whole bodies to small fragments, were taken to Dover Air Force Base in Delaware where they were sampled for DNA testing. From the 2000 containers of evidence recovered, 938 samples were collected for DNA analysis. STR typing was then performed at the Armed Forces DNA Identification Laboratory in Rockville, Maryland using Profiler Plus. Reference samples included 49 direct reference bloodstain cards from the Armed Forces DNA Repository and 348 family references.

Over the course of the next two months (September 17, 2001 to November 15, 2001), 177 identifications were made using DNA only or a combination of DNA, dental records, and fingerprints. In addition, one victim was identified solely with dental records. Unfortunately, no biological material was recovered from the crash site for five of the missing individuals (Edson et al. 2004). In addition, there were five male STR profiles that did not match reference samples from any of the victims. These samples were classified as belonging to the terrorists who hijacked the plane and were further confirmed using mitochondrial DNA testing and comparison to Near Eastern mtDNA haplotypes (Edson et al. 2004).

The Somerset County (Shanksville, Pennsylvania) Site

Shortly after departing Newark, New Jersey on the morning of September 11, 2001 United Airlines Flight 93 was hijacked by four terrorists. The plane crashed near Shanksville, Pennsylvania at 10:10 a.m. killing all 40 passengers and crew along with the four terrorists. The 1319 total remains recovered from the Somerset County crash site were all highly fragmented because the plane went straight into the ground at a speed of more than 500 miles per hour. Scientists from the Armed Forces DNA Identification Laboratory collected 592 samples for DNA analysis including 423 bones, 141 tissues, 23 teeth, 2 hairs, and 3 fingernails. Reference samples used included 55 family references and 50 direct references.

All 40 passengers and crewmembers were able to be identified through DNA alone or a combination of DNA, dental records, and fingerprints. In addition, four male STR profiles that did not match family references were observed and ascribed to the four terrorists. These samples were tested with mtDNA and found to be associated with Near Eastern mtDNA haplotypes (Edson et al. 2004).

The World Trade Center DNA Identification Effort

The DNA identification efforts for the World Trade Center (WTC) victims have become arguably the world's largest forensic case to date (Biesecker et al. 2005, Brenner & Weir 2003, Budimlija et al. 2003, Leclair et al. 2007). In the first few months, almost 20,000 pieces of human remains were collected from a pile of rubble weighing over a million tons and reaching more than 70 feet in height following the crushing collapse of the Twin Towers. The initial removal and sorting of human remains took place between September 2001 and May 2002. However, the primary DNA identification efforts went on for more than three years—almost two-and-a-half years after the last piece of debris had been removed from the WTC site.

In the end, more than 1600 victims were identified of the 2749 present when the Twin Towers collapsed. Without the capabilities of DNA testing, there would have been only a fraction of the victims identified based on other modalities such as fingerprints and dental records (Table 9.1). As of June 4, 2010, a total of 12,769 remains (59%) from a total of 21,802 recovered at the World Trade Center site have been identified and associated with one of the 1626 victims identified so far (Mecki Prinz, NYC OCME, personal communication).

Biological samples recovered from the WTC site had been subjected to extreme environmental conditions with the building collapse and subterranean fires that continued to burn in the rubble pile for three months following the terrorist attack. The jet fuel from both planes that rammed the WTC towers ignited office furnishings leading to fires throughout the buildings. The buildings collapsed due to a synergy of the damage caused by the aircraft

TABLE 9.1 Summary of World Trade Center Victim and Remains Identification Efforts Completed as of June 2004. At That Time, the Number of Victims Stood at 2749 of Which 1558 Had Been Identified. Six Years Later (June 2010), with Some Additional Bone Fragments Located and Further DNA Testing, the Number Identified Had Climbed to 1626 Based on a Total of 21,802 Remains Recovered. Although DNA Far Outstrips Other Methods in Terms of Success at Recovering Information in This Disaster, Other Modalities Were Useful in Identifying Victims or Sets of Remains. Initial Information Courtesy of Dr. Robert Shaler and More Recent Information Courtesy of Dr. Mecki Prinz, New York City Office of Chief Medical Examiner.

Modality	Victims Identified			Remains Identified		
	Single Modality	Multiple Modalities	Total IDs	Single Modality	Multiple Modalities	Total IDs
DNA	817	465	1282	4231	3685	7916
Photo	11	14	25	11	14	25
Viewed	12	2	14	12	2	14
Body X-ray	0	3	3	0	4	4
Dental	102	424	526	117	497	614
Prints	53	215	268	56	240	296
Tattoos	0	6	6	0	6	6
Personal effects	16	59	75	18	61	79
Other	7	34	41	7	101	108

impact, the removal of fire-resistive insulation from the steel support structures, and fire-induced deformation of the remaining structure (see http://wtc.nist.gov). Thus, human remains were often comingled, very fragmented, and in many cases likely vaporized.

Dr. Robert Shaler, the director of Forensic Biology at the New York City's Office of Chief Medical Examiner (NYC OCME) stood at the helm of the WTC DNA efforts throughout the many months of this investigation (Shaler 2005). Dr. Shaler and his dedicated staff coordinated the efforts and assistance of outside companies and consultants. They worked tirelessly to collate every piece of possible information in the complex process of making a genetic match between a victim's DNA profile and that of a reference sample in the form of a personal effect or a victim's biological relative. A computer program named DNA-VIEW written by Dr. Charles Brenner played an important part in determining many of these kinship identifications.

Several innovations came out of the 9/11 attacks (Vastag 2002). These included new extraction methods from bone (Holland et al. 2003), reduced size amplicons or miniSTRs (Butler et al. 2003, Holland et al. 2003), panels of single nucleotide polymorphisms (SNPs), and high-throughput mitochondrial DNA sequencing. In addition, new software was developed to aid in matching reference samples and recovered remains as well as associating remains with the same DNA profile.

Three different software programs were used extensively in the WTC victim identification efforts: Mass Fatality Identification System (M-FISys), DNA-VIEW, and Mass Disaster Kinship Analysis Program (MDKAP). M-FISys was developed by Gene Codes Forensics (Ann Arbor, MI). It used a direct match algorithm and helped in collapsing and sorting data sets to obtain identifications (Hennessey 2002). DNA-VIEW deduced kinship by pedigree analyses (Brenner & Weir 2003), while MDKAP performed kinship analyses through pair-wise comparisons (Leclair et al. 2007). A variation of the MDKAP is now commercially available as the Bloodhound program (see http://www.ananomouse.com).

One of the largest challenges from this investigation was review of the massive amounts of data produced by contracting laboratories. The flow of DNA samples and data between the various laboratories involved in this tremendous effort is illustrated in Figure 9.5. In the initial 2001 to 2005 effort, more than 52,528 STR profiles, 16,938 SNP profiles, and 31,155 mtDNA sequences were generated in an effort to identify the 2749 victims of the World Trade Center collapse based on 19,917 recovered remains—truly a heroic effort. (Note that more remains have been recovered following the initial identification effort and a few more identifications made.)

Most of the data from the recovered remains contained only partial DNA profiles making it even more difficult to sort through and piece together sufficient information to make a reliable identification. The process for assigning identities to specific remains has three stages: collapsing, screening, and testing (Brenner & Weir 2003). Clustering algorithms can be used to help reconstruct likely pedigree information when multiple family members are part of the sample set being investigated (Cowell & Mostad 2003).

Kinship and Data Analysis Panel (KADAP)

The National Institute of Justice (NIJ) organized a panel of experts, referred to as the Kinship and Data Analysis Panel (KADAP), that convened almost every other month in the

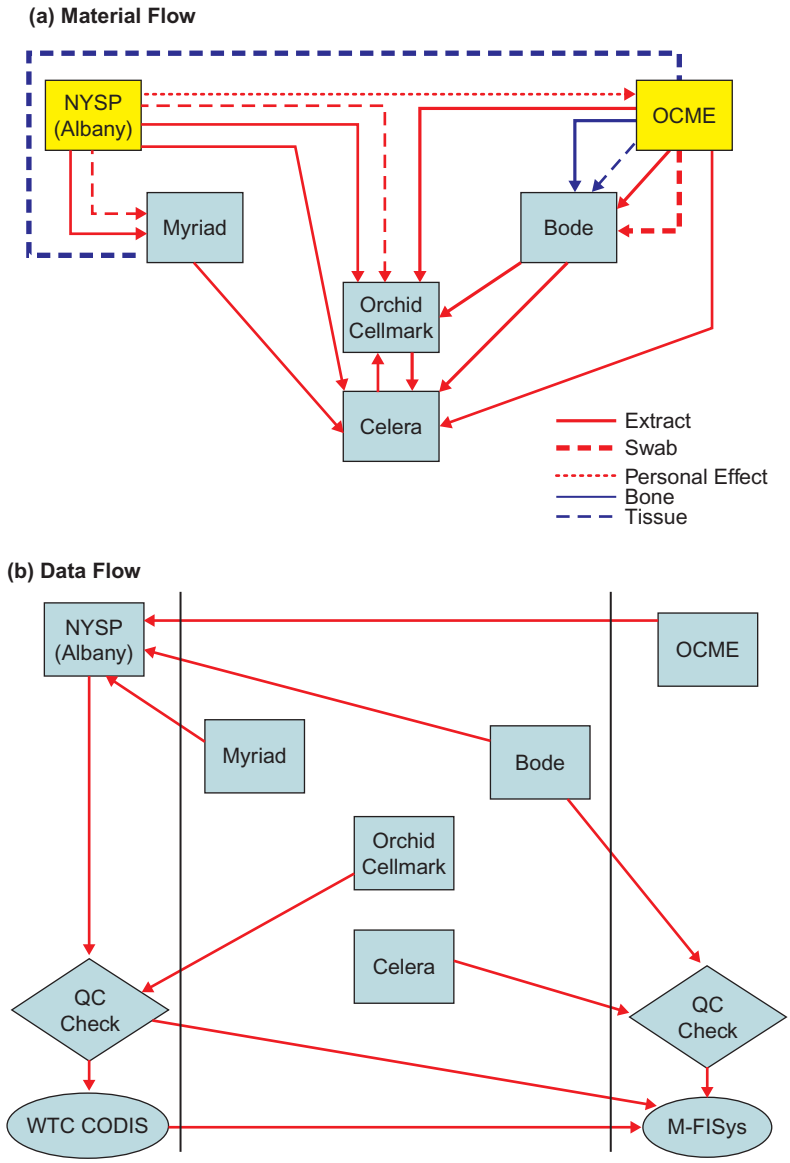


FIGURE 9.5 Illustration of (a) material and (b) data flow between laboratories involved in processing World Trade Center samples. Laboratories included Office of Chief Medical Examiner (OCME, New York City), New York State Police (NYSP, Albany, NY), Myriad Genetics (Salt Lake City, UT), Orchid Cellmark (Dallas, TX), Celera Genomics (Rockville, MD), and Bode Technology Group (Springfield, VA). Blue box colors represent the commercial labs and yellow boxes highlight the government labs. Physical materials shipped between laboratories included DNA extracts (solid red line), buccal swabs from biological relative reference samples (dashed red line), personal effects (dotted red line), recovered bones (solid blue line), and recovered tissue (dashed blue line). Note that most of the DNA samples were extracted at the NYSP and OCME laboratories although some tissue and bone were shipped directly to Myriad and Bode. All bones were extracted at Bode. Figure courtesy of National Institute of Justice World Trade Center Kinship and Data Analysis Panel and New York City Office of Chief Medical Examiner.

TABLE 9.2 World Trade Center Kinship and Data Analysis Panel (WTC KADAP) Members Convened to Aid New York City with WTC Identifications. National Institute of Justice Funded-Meetings Were Held Bimonthly from October 2001 to July 2003 in New York City, Albany, Baltimore, or Washington, DC (Not All Members Attended Every Meeting) to Discuss Progress with the Investigation and to Aid in Reviewing Technology Validation and Statistical Genetic Issues.

Member	Affiliation
Joan Bailey-Wilson, Ph.D.	National Institutes of Health-National Human Genome Research Institute
Jack Ballantyne, Ph.D.	University of Central Florida
Howard Baum, Ph.D.	New York City Office of Chief Medical Examiner
Fred Bieber, M.D, Ph.D.	Harvard Medical School
Erik Bieschke	New York City Office of Chief Medical Examiner
Les Biesecker, Ph.D.	National Institutes of Health-National Human Genome Research Institute
Charles Brenner, Ph.D.	DNA-View
Bruce Budowle, Ph.D.	Federal Bureau of Investigation Laboratory
John Butler, Ph.D.	National Institute of Standards and Technology
George Carmody, Ph.D.	Carleton University
Maureen Casey	New York City Police Department
Cheryl Conley	Orchid Genescreen Ohio
Michael Conneally, Ph.D.	Indiana University School of Medicine
Mark Dale	New York State Police; New York City Police Department Laboratory
Barry Duceman, Ph.D.	New York State Police
Art Eisenberg, Ph.D.	University of North Texas Health Science Center
Lisa Forman, Ph.D.	National Institute of Justice
Dennis Gaige	New York State Police
Mike Hennessy, MBA	Gene Codes Forensics
Steven Hogan, J.D.	New York State Police
Ken Kidd, Ph.D.	Yale University School of Medicine
Benoit Leclair, Ph.D.	Myriad Genetics
Elaine Mar	New York City Office of Chief Medical Examiner
Amy Mundorff	New York City Office of Chief Medical Examiner
Steve Niezgoda, MBA	National Institute of Justice Contractor
Judy Nolan, Ph.D.	Gene Codes Forensics
Tom Parsons, Ph.D.	Armed Forces DNA Identification Laboratory
Mecki Prinz, Ph.D.	New York City Office of Chief Medical Examiner
Elizabeth Pugh, Ph.D.	John Hopkins University/Center for Inherited Disease Research

(*Continued*)

TABLE 9.2 World Trade Center Kinship and Data Analysis Panel (WTC KADAP) Members Convened to Aid New York City with WTC Identifications. National Institute of Justice Funded-Meetings Were Held Bimonthly from October 2001 to July 2003 in New York City, Albany, Baltimore, or Washington, DC (Not All Members Attended Every Meeting) to Discuss Progress with the Investigation and to Aid in Reviewing Technology Validation and Statistical Genetic Issues. (*Continued*)

Member	Affiliation
Robert Shaler, Ph.D.	New York City Office of Chief Medical Examiner
Steve Sherry, Ph.D.	National Institutes of Health-National Center for Biotechnology Information
John Snyder	New York State Police Contractor
Amanda Sozer, Ph.D.	National Institute of Justice Contractor
Amy Sutton	Gene Codes Forensics
Steve Swinton	New York State Police
Lois Tully, Ph.D.	National Institute of Justice
Noelle Umback	New York City Office of Chief Medical Examiner
Anne Walsh, Ph.D.	New York State Department of Health
Peter Wistort	New York State Police

two years following September 11, 2001. The panel's mission was to aid in reviewing data and to provide guidance and recommendations to the NYC OCME regarding statistical thresholds for reporting DNA identifications based on direct matches and kinship associations (Biesecker et al. 2005). The names and affiliations of the advisory WTC KADAP are listed in Table 9.2. KADAP gathered in NYC, Albany, Baltimore, and Washington, DC for two-day meetings during Oct 2001, Dec 2001, Feb 2002, Apr 2002, July 2002, Sept 2002, Jan 2003, July 2003, and June 2005.

While we hope to never see the likes of another September 11, 2001 terrorist attack, forensic DNA typing laboratories should be prepared to aid in victim identification efforts in future mass fatality incidences. Lessons learned from the WTC DNA identification efforts by the KADAP were published in September 2006 by the National Institute of Justice (NIJ 2006). This report, entitled "Lessons Learned from 9/11: DNA Identification in Mass Fatality Incidents," provides information on project management, sample tracking and flow of information, suggestions on structuring communications between victim family members and the DNA laboratory, and vital issues to consider immediately after an incident occurs. It is safe to conclude that DNA analysis will continue to play a valuable support role in future mass fatality incidents, just as it does with law enforcement and the criminal justice system.

IDENTIFICATION OF HUMAN REMAINS IN MASS GRAVES USING DNA

DNA can play an important role in identifying remains in unmarked graves (Huffine 2001, Primorac 2004, Williams & Crew 2003). Unfortunately, there are regions of the world

that have suffered severely under the hands of ruthless dictators who do not value human life. Mass graves are often the tragic trademark of such tyrants.

The former Yugoslavia contains an estimated 40,000 unidentified bodies in mass graves. The International Commission on Missing Persons (ICMP) was created in 1996 to help identify human remains in these mass graves. In effect, the ICMP is using DNA technology to map human genocide. More recently hundreds of individual remains have been recovered from mass graves found in Kuwait and Iraq. In the first five years of attempting DNA analysis on recovered remains from these mass graves, 233 victims were identified (Alenizi et al. 2009).

One of the major challenges to performing DNA identification from mass graves is obtaining biological reference samples from relatives. Family outreach centers take information and blood samples from living relatives such as parents or a spouse and a child of a missing loved one. These reference samples are then typed with STRs and mitochondrial DNA sequencing to perform kinship analysis using the DNA results obtained from the remains in mass graves.

In the past, mitochondrial DNA (see Chapter 14) has often been the only source of successful DNA recovery from bones that have been in the ground for many years. However, more recently improved DNA extraction methods have enabled successful STR typing results to be obtained (e.g., Davoren et al. 2007). DNA often provides the only way to confirm the death of a missing person, enable return of the remains to a living relative, and help bring justice to the criminals who initiated the massacres that led to the mass grave sites.

READING LIST AND INTERNET RESOURCES

Missing Persons Investigations

Baker, L. E., & Backer, E. J. (2008). Reuniting families: An online database to aid in the identification of undocumented immigrant remains. *Journal of Forensic Sciences*, *53*, 50–53.

FBI national missing persons program. <http://www.fbi.gov/wanted/kidnap/kidmiss.htm>.

Glod, M. (July 21, 2009). New path to restore identities of missing: Web site combines details of remains, disappearances. *Washington Post*. Available at <http://www.washingtonpost.com/>.

Holland, M. M., et al. (1993). Mitochondrial DNA sequence analysis of human skeletal remains: Identification of remains from the Vietnam War. *Journal of Forensic Sciences*, *38*, 542–553.

James, M., et al. (2008). Missing persons in Australia. *Trends and Issues in Crime and Criminal Justice*, *353*, 1–6. Available at <http://www.aic.gov.au>.

Lorente, J. A., et al. (2002). Social benefits of non-criminal genetic databases: Missing persons and human remains identification. *International Journal of Legal Medicine*, *116*, 187–190.

National Center for Missing & Exploited Children. <http://www.missingkids.com/>.

National Institute of Justice articles. <http://www.dna.gov/identifying-persons/missing-persons/>.

National Missing and Unidentified Persons System (NamUs). <http://www.namus.gov>.

National Crime Information Center (NCIC). <http://www.fbi.gov/hq/cjisd/ncic.htm>.

Pearsall, B., & Weiss, D. M. (2010). Solving missing persons cases. *NIJ Journal*, *264*, 4–8. Available at <http://www.ncjrs.gov/pdffiles1/nij/228382.pdf>.

Piccinini, A., et al. (2010). World War One Italian and Austrian soldier identification project: DNA results of the first case. *Forensic Science International: Genetics*, *4*, 329–333.

Ritter, N. (2007). Missing persons and unidentified remains: The nation's silent mass disaster. *NIJ Journal*, *256*, 2–7. Available at <http://www.ojp.usdoj.gov/nij/journals/256/missing-persons.html>.

Sweet, D., et al. (1999). Identification of a skeleton using DNA from teeth and a PAP smear. *Journal of Forensic Sciences*, *44*, 630–633.

Torwalt, C., et al. (2005). Cervical smears as an alternative source of DNA in the identification of human skeletal remains. *Canadian Society of Forensic Sciences Journal*, *38*, 165–169.

University of North Texas Center for Human Identification. <http://www.unthumanid.org>.
Wickenheiser, R. A., et al. (1999). Case of identification of severely burned human remains via paternity testing with PCR DNA typing. *Canadian Society of Forensic Sciences Journal*, *32*, 15–24.

Disaster Victim Identification (DVI)

Alonso, A., et al. (2005). Challenges of DNA profiling in mass disaster investigations. *Croatian Medical Journal*, *46*, 540–548.
Buckleton, J., et al. (2005). Disaster victim identification, identification of missing persons, and immigration cases. In J. Buckleton, C. M. Triggs, T. Clayton (Eds.), *Forensic DNA evidence interpretation* (Chapter 11, pp. 395–437). Boca Raton: CRC Press.
Corach, D., et al. (1995). Mass disasters: Rapid molecular screening of human remains by means of short tandem repeats typing. *Electrophoresis*, *16*, 1617–1623.
De Valck, E. (2006). Major incident response: Collecting antemortem data. *Forensic Science International*, *159*(Suppl.), S15–S19.
Graham, E. A. M. (2006). Disaster victim identification. *Forensic Science, Medicine, and Pathology*, *2*, 203–207.
Li, T. H., et al. (2006). Interpreting anonymous DNA samples from mass disasters – probabilistic forensic inference using genetic markers. *Bioinformatics*, *22*(14), e298–e306.
SNA International. <http://www.sna-intl.com/>.
Winskog, C., et al. (2010). The use of commercial alcohol products to sterilize bones prior to DNA sampling. *Forensic Science, Medicine, and Pathology*, *6*, 127–129.

DVI Recommendations

AABB. (2010). *Guidelines for mass fatality DNA identification operations*. Available at <http://www.aabb.org/programs/disasterresponse/Documents/aabb>.
Bradford, L., et al. (2010). Disaster victim investigation recommendations from two simulated mass disaster scenarios utilized for user acceptance testing CODIS 6.0. *Forensic Science International: Genetics* (in press). doi:10.1016/j.fsigen.2010.05.005
Budowle, B., et al. (2005). Forensic aspects of mass disasters: Strategic considerations for DNA-based human identification. *Legal Medicine*, *7*, 230–243.
International Criminal Police Organization. (2008). *Interpol disaster victim identification guide*. Available at <http://www.interpol.int/Public/DisasterVictim/default.asp>.
International Committee of the Red Cross. (2009). *Missing people, DNA analysis and identification of human remains: A guide to best practice in armed conflicts and other situations of armed violence*. 2nd ed. Available at <http://www.icrc.org/Web/Eng/siteeng0.nsf/htmlall/p4010/$File/ICRC_002_4010.PDF>.
Lee, J., et al. (2008). Recommendations for DNA laboratories supporting disaster victim identification (DVI) operations—Australian and New Zealand consensus on ISFG recommendations. *Forensic Science International: Genetics*, *3*, 54–56.
Lessig, R., et al. (2011). German standards for forensic molecular genetics investigations in cases of mass disaster victim identification (DVI). *Forensic Science International: Genetics*, *5*, 247–248.
National Institute of Justice. (2005). *Mass fatality incidents: A guide for human forensic identification*. <http://www.ojp.usdoj.gov/nij/pubs-sum/199758.htm>.
National Institute of Justice. (2006). *Lessons learned from 9/11: DNA identifications in mass fatality incidents*. <http://massfatality.dna.gov>.
Prinz, M., et al. (2007). DNA Commission of the International Society of Forensic Genetics (ISFG): Recommendations regarding the role of forensic genetics for disaster victim identification (DVI). *Forensic Science International: Genetics*, *1*, 3–12.
Sweet, D. (2010). INTERPOL DVI best-practice standards—an overview. *Forensic Science International*, *201*, 18–21.
Westen, A. A., et al. (2008). Femur, rib, and tooth sample collection for DNA analysis in disaster victim identification (DVI): A method to minimize contamination risk. *Forensic Science Medicine, and Pathology*, *4*, 15–21.

Waco Branch Davidian Fire

Clayton, T. M., et al. (1995). Identification of bodies from the scene of a mass disaster using DNA amplification and short tandem repeat (STR) loci. *Forensic Science International*, *76*, 7–15.

DiZinno, J., et al. (1994). The Waco, Texas incident: The use of DNA analysis to identify human remains. *Proceedings from the fifth international symposium on human identification* (pp. 129–135). Madison, Wisconsin: Promega.

Airplane Crashes

Ballantyne, J. (1997). Mass disaster genetics. *Nature Genetics*, *15*, 329–331.

Goodwin, W., et al. (1999). The use of mitochondrial DNA and short tandem repeat typing in the identification of air crash victims. *Electrophoresis*, *20*, 1707–1711.

Hsu, C. M., et al. (1999). Identification of victims of the 1998 Taoyuan Airbus crash accident using DNA analysis. *International Journal of Legal Medicine*, *113*, 43–46.

Olaisen, B., et al. (1997). Identification by DNA analysis of the victims of the August 1996 Spitsbergen civil aircraft disaster. *Nature Genetics*, *15*, 402–405.

Swissair Flight 111

Frégeau, C. J., et al. (2000). The Swissair Flight 111 disaster: Short tandem repeat mutations observed. *Progress in Forensic Genetics*, *8*, 40–42.

Leclair, B., et al. (2000). Enhanced kinship analysis and STR-based DNA typing for human identification in mass disasters. *Progress in Forensic Genetics*, *8*, 91–93.

Leclair, B., et al. (2004). Enhanced kinship analysis and STR-based DNA typing for human identification in mass fatality incidents: The Swissair flight 111 disaster. *Journal of Forensic Sciences*, *49*, 939–953.

Southeast Asian Tsunami

Brenner, C. H. (2006). Some mathematical problems in the DNA identification of victims in the 2004 tsunami and similar mass fatalities. *Forensic Science International*, *157*, 172–180.

Deng, Y. -J., et al. (2005). Preliminary DNA identification for the tsunami victims in Thailand. *Genomics, Proteomics, and Bioinfomatics*, *3*, 143–157.

Lau, G., et al. (2005). After the Indian Ocean tsunami: Singapore's contribution to the international disaster victim identification effort in Thailand. *Annals of the Academy of Medicine Singapore*, *34*(5), 341–351.

Morgan, O. W., et al. (2006). Mass fatality management following the South Asian tsunami disaster: Case studies in Thailand, Indonesia, and Sri Lanka. *PLoS Medicine*, *3*(6), e195.

Zehner, R. (2007). "Foreign" DNA in tissue adherent to compact bone from tsunami victims. *Forensic Science International: Genetics*, *1*, 218–222.

9/11/01: Pentagon and PA Sites

Edson, S. M., et al. (2004). Naming the dead—confronting the realities of rapid identification of degraded skeletal remains. *Forensic Science Review*, *16*, 63–90.

9/11/01: World Trade Center DNA Identification Effort

Biesecker, L. G., et al. (2005). DNA identifications after the 9/11 World Trade Center attack. *Science*, *310*, 1122–1123.

Brenner, C. H., & Weir, B. S. (2003). Issues and strategies in the identification of World Trade Center victims. *Theoretical and Population Biology*, *63*, 173–178.

Budimlija, Z. M., et al. (2003). World Trade Center human identification project: Experiences with individual body indentification cases. *Croatian Medical Journal*, *44*, 259–263.

Cowell, R. G., & Mostad, P. (2003). A clustering algorithm using DNA marker information for sub-pedigree reconstruction. *Journal of Forensic Sciences*, *48*, 1239–1248.

Gill, J. R. (2006). 9/11 and the New York City Office of Chief Medical Examiner. *Forensic Science, Medicine, and Pathology*, *2*, 29–32.

Hennessey, M.. (2002). World Trade Center DNA identifications: The administrative review process. *Proceedings of the Thirteenth International Symposium on Human Identification*. Available at <http://www.promega.com/geneticidproc/ussymp13proc/contents/hennesseyrev1.pdf>.

Holland, M. M., et al. (2003). Development of a quality, high throughput DNA analysis procedure for skeletal samples to assist with the identification of victims from the World Trade Center attacks. *Croatian Medical Journal*, *44*, 264–272.
Leclair, B., et al. (2007). Bioinformatics and human identification in mass fatality incidents: The World Trade Center disaster. *Journal of Forensic Sciences*, *52*, 806–819.
National Institute of Justice. (2006). *Lessons learned from 9/11: DNA identifications in mass fatality incidents*. Available at <http://massfatality.dna.gov>.
Ritter, N. (2007). Identifying remains: Lessons learned from 9/11. *NIJ Journal*, *256*, 20–26. Available at <http://www.ojp.usdoj.gov/nij/journals/256/lessons-learned.html>.
Shaler, R. C. (2005). *Who they were—inside the world trade center DNA story: The unprecedented effort to identify the missing*. New York: Free Press.
Vastag, B. (2002). Out of tragedy, identification innovation. *JAMA: The Journal of the American Medical Association*, *288*, 1221–1223.

Hurricane Katrina

Dolan, S. M., et al. (2009). The emerging role of genetics professionals in forensic kinship DNA identification following a mass fatality: Lessons learned from Hurricane Katrina volunteers. *Genetics in Medicine*, *11*, 414–417.
Donkervoort, S., et al. (2008). Enhancing accurate data collection in mass fatality kinship identifications: Lessons learned from Hurricane Katrina. *Forensic Science International: Genetics*, *2*, 354–362.

Other

Hartman, D., et al. (2011). The 2009 Victorian Bushfires Disaster: The importance of Guthrie cards and other medical samples for the direct matching of disaster victims using DNA profiling. *Forensic Science International, 205,* 59–63.
Lin, C. -Y., et al. (2010). The strategies to DVI challenges in Typhoon Morakot. *International Journal of Legal Medicine* (in press) (doi:10.1007/s00414-010-0479-8).
Sudoyo, H., et al. (2008). DNA analysis in perpetrator identification of terrorism-related disaster: Suicide bombing of the Australian Embassy in Jakarta 2004. *Forensic Science International: Genetics*, *2*, 231–237.

Mass Graves

Alenizi, M. A., et al. (2009). Concordance between the AmpFlSTR MiniFiler and AmpFlSTR Identifiler PCR Amplification kits in the Kuwaiti population. *Journal of Forensic Sciences*, *54*, 350–352.
Baraybar, J. P. (2008). When DNA is not available, can we still identify people? Recommendations for best practice. *Journal of Forensic Sciences*, *53*, 533–540.
Biruš, I., et al. (2003). How high should paternity index be for reliable identification of war victims by DNA typing? *Croatian Medical Journal*, *44*, 322–326.
Cox, M., et al. (2008). *The scientific investigation of mass graves: Towards protocols and standard operating procedures*. New York: Cambridge University Press.
Davoren, J., et al. (2007). Highly effective DNA extraction method for nuclear short tandem repeat testing of skeletal remains from mass graves. *Croatian Medical Journal*, *48*, 478–485.
Gornik, I., et al. (2002). The identification of war victims by reverse paternity is associated with significant risks of false inclusion. *International Journal of Legal Medicine*, *116*, 255–257.
Huffine, E., et al. (2001). Mass identification of persons missing from the break-up of the former Yugoslavia: Structure, function, and role of the International Commission on Missing Persons. *Croatian Medical Journal*, *42*, 271–275.
Huffine, E., et al. (2007). Developing role of forensics in deterring violence and genocide. *Croatian Medical Journal*, *48*, 431–436.
International Commission on Missing Persons (ICMP). <http://www.ic-mp.org>.
Kračun, S. K., et al. (2007). Population substructure can significantly affect reliability of a DNA-led process of identification of mass fatality victims. *Journal of Forensic Sciences*, *52*, 874–878.
Primorac, D. (2004). The role of DNA technology in identification of skeletal remains discovered in mass graves. *Forensic Science International*, *146S*, S163–S164.
Williams, E. D., & Crews, J. D. (2003). From dust to dust: Ethical and practical issues involved in the location, exhumation, and identification of bodies from mass graves. *Croatian Medical Journal*, *44*, 251–258.